

OLLAMA

Complete Model Library & Platform Reference

CachyOS · Pixel 9 / Android · Termux | March 2026

This manual catalogs every model in the Ollama library, explains each model's capabilities, provides installation and configuration guidance for CachyOS (KDE Plasma, GPU-accelerated) and Pixel 9 Android (Termux, CPU inference), and includes platform-specific recommendations for every use case.

CachyOS / Desktop

Pixel 9 / Android

Ollama Platform Overview

1.1 What Is Ollama?

Ollama is an open-source local LLM runtime that downloads, manages, and serves large language models directly on your hardware. It wraps llama.cpp internals behind a clean CLI and an OpenAI-compatible REST API, abstracting GPU acceleration, GGUF quantization, and context management so you focus on using models rather than configuring them.

Every model in the Ollama library is distributed as GGUF files (or sharded files for very large models). Ollama handles download, caching, hot-loading, and graceful unloading automatically.

1.2 Architecture

ollama serve — The persistent background daemon. Listens on `127.0.0.1:11434` by default. Manages GPU/CPU layer allocation, KV-cache, and model slots.

ollama pull <model> — Downloads a model from the Ollama registry into the local model store (`~/.ollama/models` or `/var/lib/ollama/.ollama/models` when run as a service).

ollama run <model> — Starts an interactive REPL. Pulls the model first if not already downloaded.

REST API — `POST /v1/chat/completions` for OpenAI-compatible chat. `POST /api/generate` for raw completion. `POST /api/embeddings` for vector embeddings.

1.3 Quantization Guide

Every model is available in multiple quantization variants. Choosing the right one determines model quality, VRAM usage, and inference speed.

Tag	Bits/Weight	~Size (7B model)	When to Use
Q2_K	2-bit	~2.7 GB	Extreme RAM limits only — noticeable quality loss
Q3_K_M	3-bit	~3.3 GB	Very tight RAM; usable for simple tasks
Q4_K_M	4-bit	~4.1 GB	Best default — excellent quality/size ratio. Ollama's standard.
Q5_K_M	5-bit	~4.8 GB	Step up when VRAM allows; near-original quality
Q6_K	6-bit	~5.5 GB	High quality, moderate VRAM. Ideal for CachyOS with 12GB+ VRAM.
Q8_0	8-bit	~7.2 GB	Near-lossless; use when VRAM is plentiful

 **Tip:** Ollama defaults to Q4_K_M for most models. Request specific quants by appending the tag: `ollama run llama3.1:8b-instruct-q5_K_M`

1.4 Model Tag Glossary

- tools** — Supports function/tool calling for agentic workflows
- thinking** — Chain-of-thought reasoning mode; model shows reasoning steps before answering

- **vision** — Accepts image input alongside text
- **embedding** — Produces vector embeddings for RAG/search; not a chat model
- **cloud** — Model is very large; typically requires cloud runner or 64GB+ RAM hardware

1.5 Hardware Requirements

Model Size	Min RAM/VRAM	Recommended	Example Models
1B–3B	4 GB	8 GB	TinyLlama, SmolLM2, Llama 3.2 1B/3B, DeepSeek-R1 1.5B
7B–8B	8 GB	16 GB	Llama 3.1 8B, Mistral 7B, Gemma2 9B, Qwen2.5 7B
13B–14B	16 GB	24 GB	Phi-4 14B, CodeLlama 13B, Qwen2.5 14B
22B–32B	24 GB	32 GB	Mistral Small 3, QwQ 32B, DeepSeek-R1 32B
70B	40 GB	48–64 GB	Llama 3.3 70B, DeepSeek-R1 70B
405B–671B	128 GB+	256 GB	Llama 3.1 405B, DeepSeek-V3 671B

2. Platform Setup

2.1 CachyOS Installation

CachyOS carries Ollama in its optimized repositories, rebuilt against x86-64-v3/v4/znver4 instruction sets for automatic AVX2/AVX-512 uplift. No AUR helper needed.

```
# CPU-only
sudo pacman -S ollama
# NVIDIA CUDA (recommended if you have an NVIDIA GPU)
sudo pacman -S ollama ollama-cuda
# AMD ROCm
sudo pacman -S ollama ollama-rocm
# Vulkan (experimental)
sudo pacman -S ollama ollama-vulkan

# Enable and start the systemd service
sudo systemctl enable --now ollama.service
sudo systemctl status ollama.service
```

 **CachyOS Note:** The CachyOS-optimized Ollama build (currently v0.17.5-1.1) automatically uses your architecture-specific instruction set. The BORE scheduler and kernel PGO optimizations give measurable throughput improvements over generic distros.

Performance Tuning (CachyOS)

```
sudo systemctl edit ollama.service
# Add inside the override file:
[Service]
Environment="OLLAMA_HOST=0.0.0.0:11434"
Environment="OLLAMA_FLASH_ATTENTION=1"
Environment="OLLAMA_KV_CACHE_TYPE=q8_0"
Environment="OLLAMA_MAX_LOADED_MODELS=2"
Environment="OLLAMA_NUM_PARALLEL=4"

sudo systemctl daemon-reload && sudo systemctl restart ollama.service
```

- `OLLAMA_FLASH_ATTENTION=1` — Reduces VRAM usage significantly (NVIDIA CC $\geq$ 7.0, AMD RDNA2+)
- `OLLAMA_KV_CACHE_TYPE=q8_0` — Quantizes KV cache to 8-bit, saves VRAM with minimal quality impact
- `OLLAMA_MAX_LOADED_MODELS=2` — Keep two models hot in VRAM simultaneously
- `OLLAMA_NUM_PARALLEL=4` — Allow 4 concurrent inference slots when using as a local API server

2.2 Pixel 9 / Android via Termux

Ollama is available directly in the Termux package repository (as of April 2025). No PRoot layer required for basic usage.

 **Android Limitation:** Ollama on Android runs CPU-only. The Tensor G4 GPU and NPU are NOT utilized by Ollama. Keep models at 1B–3B for usable inference speeds on the Pixel 9.

Method A: Native Termux (Simplest)

```
# Install Termux from F-Droid (NOT Play Store — outdated there)
pkg update && pkg upgrade
pkg install ollama zellij
```

```
# Disable battery optimization for Termux in Android settings
# Developer Options > Feature flags > disable_child_process_killer

# Start server in background session
ollama serve &

# Pull and run a Pixel-safe model
ollama pull llama3.2:3b
ollama run llama3.2:3b
```

Pixel 9 Environment Settings

```
# Add to ~/.bashrc in Termux:
export OLLAMA_NUM_PARALLEL=1
export OLLAMA_MAX_LOADED_MODELS=1
export OLLAMA_KEEP_ALIVE=10m
alias ols='ollama serve &'
alias olr='ollama run'
```

 **Speed Expectation:** CPU inference on Pixel 9 yields roughly 3–8 tokens/sec for 1B models, and 1–3 tokens/sec for 3B. These are best used for offline assistants, not real-time coding help.

3. Complete Model Library

All 100+ models in the Ollama library as of March 2026. Models are grouped by category. Green cells = recommended CachyOS variant. Purple cells = recommended Pixel 9 variant. — = not suitable for that platform.

 **Reading the table:** Pull counts show library popularity. Sizes shows available parameter counts. CachyOS Pick / Pixel 9 Pick recommends the specific size:quantization tag to use.

3.1 General Purpose / Flagship Models

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
llama3.1	8B/70B/405B	Tools, Long Ctx	Meta flagship. 110.6M pulls — #1 most downloaded. 8B excels at general reasoning, coding, instruction following. 128K context. 70B is enterprise-grade.	70B:Q5_K_M	llama3.2:1b alt
llama3.2	1B / 3B	Tools, Mobile	Meta's compact on-device series. Tool calling in both sizes. 1B runs instantly on anything; 3B is the Pixel 9 sweet spot.	3B:Q8_0	3B:Q4_K_M ★
llama3.3	70B	Tools	New SOTA 70B from Meta. Comparable to Llama 3.1 405B at a fraction of compute. Best open-weight 70B in late 2025.	70B:Q4_K_M	—
llama3	8B/70B	General	Original Llama 3. Superseded by 3.1/3.2/3.3 but stable and well-supported.	8B:Q5_K_M	—
llama2	7B/13B/70B	General	Second-gen Meta model. Largely superseded but very stable for offline archival setups.	13B:Q4_K_M	—
mistral	7B	Tools	Mistral 7B v0.3. Fast, punches above its weight. One of the best 7B models available.	7B:Q6_K	—

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
mistral-nemo	12B	Tools, 128K ctx	12B co-developed with NVIDIA. 128K context window.	12B:Q4_K_M	—
mistral-small	22B/24B	Tools	Mistral Small 3 sets benchmarks for sub-70B models.	24B:Q4_K_M	—
mistral-large	123B	Tools, Code, Math	Mistral Large 2. Flagship with 128K context, multi-language, strong reasoning.	Needs 64GB+	—
gemma3	270M–27B	Vision, Multi-img	Google's most capable single-GPU family. Gemini-class locally. Multi-image input. 4B is a standout performer.	12B:Q5_K_M	4B:Q4_K_M ★
gemma2	2B/9B/27B	General	Google Gemma 2. Flash attention enabled by default in Ollama.	9B:Q6_K	2B:Q4_K_M
gemma	2B/7B	General	Original Gemma. Superseded by Gemma2/3 but lightweight.	7B:Q5_K_M	2B:Q4_K_M
phi4	14B	Reasoning, Code	Microsoft Phi-4. SOTA 14B, exceptional reasoning and coding. High-quality synthetic training data.	14B:Q5_K_M	—
phi4-mini	3.8B	Tools, Math, Multilingual	Compact Phi-4 with tool calling. Strong math and multilingual in under 4B params.	3.8B:Q8_0	3.8B:Q4_K_M ★
phi4-reasoning	14B	Reasoning, Math	Phi-4 fine-tuned for step-by-step reasoning. Rivals much larger models on math benchmarks.	14B:Q4_K_M	—
phi3	3.8B/14B	General	Microsoft Phi-3 Mini/Medium. Phi-3 Mini punches above its weight on MMLU.	14B:Q4_K_M	3.8B:Q4_K_M ★

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
phi3.5	3.8B	General	Updated Phi-3, performance exceeds larger models.	3.8B:Q8_0	3.8B:Q4_K_M
qwen3	0.6B–235B	Tools, Thinking, MoE	Latest Qwen generation. Dense and MoE variants. Thinking mode. 8B is an excellent CachyOS daily driver.	14B:Q5_K_M	1.7B:Q4_K_M ★
qwen2.5	0.5B–72B	Tools, 128K, Multilingual	Alibaba workhorse. 18T token pretraining. 128K context. Very strong across the size range.	14B:Q5_K_M	3B:Q4_K_M
olmo2	7B/13B	General, Open	OLMo 2 trained on 5T tokens. Competitive with Llama 3.1 on academic benchmarks. Fully open.	13B:Q5_K_M	—

3.2 Reasoning & Math Models

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
deepseek-r1	1.5B–671B	Tools, Thinking	78.8M pulls — #2 most downloaded. Open reasoning family approaching O3 and Gemini 2.5 Pro. Extended chain-of-thought. 7B/8B are accessible; 671B is elite.	32B:Q4_K_M	1.5B:Q4_K_M ★
qwq	32B	Tools, Reasoning	QwQ is the Qwen reasoning model. Very strong at math and logic. Think-then-answer approach.	32B:Q4_K_M	—
phi4-mini-reasoning	3.8B	Reasoning, Math	Lightweight Phi-4 for step-by-step math reasoning	3.8B:Q8_0	3.8B:Q4_K_M ★

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
			on constrained hardware.		
deepscaler	1.5B	Math Reasoning	Fine-tuned DeepSeek-R1 distill. Surpasses o1-preview on math at just 1.5B params.	1.5B:Q8_0	1.5B:Q4_K_M★
qwen2-math	1.5B/7B/72B	Math	Specialized math model on Qwen2. Outperforms GPT-4o on math benchmarks.	7B:Q5_K_M	1.5B:Q4_K_M
mathstral	7B	Math, Science	Mistral 7B fine-tuned for math reasoning and scientific discovery.	7B:Q6_K	—
exaone-deep	2.4B/7.8B/32B	Reasoning, Math, Code	LG AI Research. 7.8B beats o1-mini. Strong math and coding benchmarks.	7.8B:Q5_K_M	2.4B:Q4_K_M★
openthinker	7B/32B	Reasoning	Fully open reasoning model distilled from DeepSeek-R1.	32B:Q4_K_M	—
magistral	24B	Tools, Thinking	Mistral's efficient reasoning model. 24B parameters.	24B:Q4_K_M	—
cogito	3B–70B	Tools, Reasoning	Deep Cogito hybrid reasoning models. Outperform best open models of same size.	14B:Q5_K_M	3B:Q4_K_M
marco-o1	7B	Reasoning	Alibaba AIDC reasoning model for real-world solutions.	7B:Q5_K_M	—

3.3 Coding Models

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
qwen2.5-coder	0.5B–32B	Tools, Code	11.4M pulls. Latest code-specific Qwen. Major improvements in generation, reasoning, and fixing across 92+ languages.	14B:Q5_K_M	3B:Q4_K_M ★
deepseek-coder-v2	16B/236B	Code (MoE)	Open-source MoE code model. GPT-4-Turbo level on code tasks. 16B is the accessible size.	16B:Q4_K_M	—
deepseek-coder	1.3B/6.7B/33B	Code	Trained on 2T code+language tokens. Strong baseline coding model.	6.7B:Q6_K	1.3B:Q4_K_M
codellama	7B/13B/34B/70B	Code	Meta's code Llama. 20+ languages. 34B handles complex debugging.	34B:Q4_K_M	—
devstral	24B	Tools, Code Agents	Mistral + All Hands AI. Best open-source model for coding agents. 46.8% SWE-Bench. 128K ctx.	24B:Q4_K_M	—
codestral	22B	Code Generation	Mistral's dedicated code model. Very strong at completion and generation.	22B:Q4_K_M	—
deepcoder	1.5B/14B	Code, Reasoning	Fully open 14B coder at O3-mini level. 1.5B accessible for constrained hardware.	14B:Q4_K_M	1.5B:Q4_K_M ★
qwen3-coder	30B/480B	Tools, Long Ctx, Agentic	Alibaba's long-context agentic coding. Handles	30B:Q4_K_M	—

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
			massive codebases.		
starcoder2	3B/7B/15B	Code	Next-gen StarCoder. Transparently trained, open weights.	15B:Q5_K_M	3B:Q4_K_M
codegemma	2B/7B	Code, FIM	Google CodeGemma. Fill-in-middle, code gen, instruction following.	7B:Q5_K_M	2B:Q4_K_M
opencoder	1.5B/8B	Code	Open reproducible code LLM. English and Chinese. 8B is strong.	8B:Q5_K_M	1.5B:Q4_K_M
yi-coder	1.5B/9B	Code	Yi code series. SOTA coding under 10B parameters.	9B:Q5_K_M	1.5B:Q4_K_M
granite-code	3B/8B/20B/34B	Code (IBM)	IBM Granite code family. Enterprise code intelligence.	20B:Q4_K_M	3B:Q4_K_M
stable-code	3B	Code (Compact)	Stable Code 3B on par with CodeLlama 7B at 2.5x smaller.	3B:Q8_0	3B:Q4_K_M ★
sqlcoder	7B/15B	SQL Code	Fine-tuned StarCoder for SQL generation. Very accurate text-to-SQL.	15B:Q4_K_M	—
codeqwen	7B	Code	CodeQwen1.5. 64K context, 92 languages, strong text-to-SQL.	7B:Q6_K	—

3.4 Vision & Multimodal Models

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
llama3.2-vision	11B/90B	Vision, Reasoning	Llama 3.2 with image input. 11B handles image reasoning, chart reading, VQA. 90B is elite.	11B:Q4_K_M	—
llava	7B/13B/34B	Vision	LLaVA 1.6. Combines vision encoder with Vicuna. Classic general-purpose VQA model.	13B:Q4_K_M	—
gemma3	270M–27B	Vision, Multi-img	Gemma 3 accepts multiple images. Advanced visual comparisons, reasoning.	12B:Q4_K_M	4B:Q4_K_M ★
qwen2.5vl	3B/7B/32B/72B	Vision, OCR	Flagship Qwen vision-language. Document OCR, layout analysis.	7B:Q5_K_M	3B:Q4_K_M
qwen3-vl	2B–235B	Vision, Tools, Thinking	Most powerful Qwen vision model. Code from visuals, UI/UX conversion.	8B:Q4_K_M	—
llava-llama3	8B	Vision	LLaVA fine-tuned from Llama 3 Instruct. Better benchmark scores.	8B:Q5_K_M	—
llava-phi3	3.8B	Vision (Compact)	Small LLaVA from Phi-3 Mini. Vision in a compact package.	3.8B:Q8_0	3.8B:Q4_K_M ★
moondream	1.8B	Vision (Edge)	Designed for CPU and mobile. The smallest usable vision model.	1.8B:Q8_0	1.8B:Q4_K_M ★
minicpm-v	8B	Vision	Multimodal LLM series for vision-language understanding.	8B:Q5_K_M	—
deepseek-ocr	3B	Vision, OCR	Token-efficient OCR from DeepSeek.	3B:Q8_0	3B:Q4_K_M

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
			Document text extraction.		
granite3.2-vision	2B	Vision, Tools, Docs	IBM compact vision model. Tables, charts, diagrams extraction.	2B:Q8_0	2B:Q4_K_M★
bakllava	7B	Vision	Mistral 7B augmented with LLaVA architecture. Multimodal.	7B:Q5_K_M	—

3.5 Embedding Models

Embedding models convert text to numerical vectors for semantic search and RAG pipelines. They do not generate text.

Model	Size	Type	Description	Best Use
nomic-embed-text	137M	Dense	55.2M pulls — the default choice for most RAG setups. Large token context window.	RAG/Search
mxbai-embed-large	335M	Dense	State-of-the-art large embedding from mixedbread.ai. Strong retrieval benchmarks.	Retrieval
bge-m3	567M	Dense	BAAI model. Multi-functional, multilingual, multi-granularity. 3.4M pulls.	Multilingual RAG
all-minilm	22M/33M	Dense	Tiny and fast. Sentence-level similarity. Good for lightweight local search.	Mobile RAG
snowflake-arctic-embed	22M–335M	Dense	Snowflake suite. Multiple sizes for different hardware.	Enterprise
qwen3-embedding	0.6B/4B/8B	Dense	Built on Qwen3. Comprehensive multilingual text embeddings.	Large-scale RAG
granite-embedding	30M/278M	Biencoder	IBM Granite embeddings. 30M English-only, 278M multilingual.	IBM Workloads

3.6 Uncensored / Dolphin Family

These models have had safety fine-tuning removed or reduced. Useful for creative writing, research, and direct outputs without guardrails.

Model	Size	Type	Description	CachyOS
dolphin3	8B	General, Code, Agentic	Dolphin 3.0 on Llama 3.1 8B. 3.6M pulls. Coding, math, function calling, general use. Ultimate general-purpose local model.	8B:Q5_K_M
dolphin-llama3	8B/70B	Instruct, Code	Dolphin 2.9 on Llama 3. Instruction, conversational, and coding skills.	8B:Q6_K
dolphin-mixtral	8x7B/8x22B	MoE, Code	Uncensored fine-tune of Mixtral MoE. Excels at coding. 1.1M pulls.	8x7B:Q4_K_M
dolphin-phi	2.7B	Compact, Uncensored	Tiny uncensored model based on Microsoft Phi. Very lightweight at 2.7B.	2.7B:Q8_0
llama2-uncensored	7B/70B	General	Uncensored Llama 2 by George Sung and Jarrad Hope.	7B:Q6_K
wizard-vicuna-uncensored	7B/13B/30B	General	Wizard Vicuna uncensored on Llama 2 by Eric Hartford. 567K pulls.	13B:Q4_K_M

3.7 Mixture-of-Experts (MoE) Models

MoE models use multiple specialized sub-networks, activating only a fraction per token — delivering high quality at lower effective compute than a dense model of equivalent total parameters.

Model	Params	Type	Description	CachyOS
deepseek-v3	671B total, 37B active	MoE, General	Strong MoE. 671B total but only 37B activated per token.	Needs 40GB+
mixtral	8x7B / 8x22B	Tools, MoE	Mistral MoE. 8x7B activates 2 experts per token (~13B effective). Very strong at coding. 1.9M pulls.	8x7B:Q5_K_M
qwen3	30B/235B MoE	Tools, Thinking	Qwen3 includes dense and MoE variants. 235B MoE with sparse activation.	30B:Q4_K_M
granite3.1-moe	1B / 3B	Tools, Low Latency	IBM tiny MoE models. Designed for low latency on edge hardware.	3B:Q8_0

3.8 Multilingual & Translation Models

Model	Size	Languages	Description	CachyOS
aya	8B/35B	23 languages	Cohere Aya 23. State-of-the-art multilingual. Covers many underserved languages.	8B:Q6_K
aya-expanse	8B/32B	23 languages	Cohere improved multilingual. Better performance across all 23 languages.	8B:Q5_K_M
translategemma	4B/12B/27B	55 languages	Gemma 3-based translation. Open translation for 55 languages.	12B:Q4_K_M
llama2-chinese	7B/13B	Chinese/English	Llama 2 fine-tuned for Chinese dialogue. Bilingual.	13B:Q4_K_M
sailor2	1B/8B/20B	SE Asian	Multilingual for South-East Asia. 1B is very accessible.	8B:Q5_K_M
glm4	9B	Multilingual	Strong multilingual model competitive with Llama 3.	9B:Q5_K_M

3.9 Compact / Edge Models (Mobile-First)

These models are designed or proven to run on mobile hardware, edge devices, and CPU-only environments like the Pixel 9 running Termux.

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
smollm2	135M/360M/1.7B	Tools (Compact)	HuggingFace SmolLM2. 135M is absurdly small yet functional. 1.7B has tool calling.	1.7B:Q8_0	1.7B:Q4_K_M ★
smollm	135M/360M/1.7B	Compact	Original SmolLM. Slightly less capable but very lightweight. 993K pulls.	1.7B:Q8_0	1.7B:Q4_K_M
tinylama	1.1B	Compact	TinyLlama trained on 3T tokens. Fast startup, tiny footprint.	1.1B:Q8_0	1.1B:Q4_K_M ★
tinydolphin	1.1B	Compact, Uncensored	Dolphin 2.8 data on	1.1B:Q8_0	1.1B:Q4_K_M

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
			TinyLlama. Uncensored tiny model.		
llama3.2	1B/3B	Tools, Mobile	Meta's intentional on-device models. Tool calling even in 1B.	3B:Q8_0	3B:Q4_K_M ★
deepscaler	1.5B	Math Reasoning	Math-focused tiny model. Beats o1-preview on math evals.	1.5B:Q8_0	1.5B:Q4_K_M ★
deepseek-r1	1.5B	Thinking	Smallest DeepSeek-R1 distill. Reasoning chain-of-thought in 1.5B.	1.5B:Q8_0	1.5B:Q4_K_M ★
granite4	350M/1B/3B	Tools, Edge	IBM Granite 4 for edge. Improved tool calling and instruction following.	3B:Q8_0	1B:Q4_K_M ★
moondream	1.8B	Vision (Edge)	Tiny vision-language for edge. Designed for CPU and mobile.	1.8B:Q8_0	1.8B:Q4_K_M ★
l1m2.5-thinking	1.2B	Tools, Thinking	Tiny hybrid model with thinking mode. On-device reasoning in 1.2B.	1.2B:Q8_0	1.2B:Q4_K_M ★
reader-lm	0.5B/1.5B	HTML→Markdown	Converts HTML to clean Markdown. Specialized	1.5B:Q8_0	0.5B:Q8_0

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
			for content pipelines.		
phi4-mini	3.8B	Tools, Math, Multilingual	Best quality near-4B for mobile. Phi-4 with function calling.	3.8B:Q8_0	3.8B:Q4_K_M★

3.10 Specialized Domain Models

Model	Sizes	Capabilities	Description	CachyOS Pick	Pixel 9 Pick
meditron	7B/70B	Medical	Open-source medical LLM from Llama 2. Clinical reasoning and medical literature.	7B:Q5_K_M	—
medllama2	7B	Medical QA	Fine-tuned Llama 2 on open medical datasets for clinical Q&A.	7B:Q5_K_M	—
sqlcoder	7B/15B	SQL	StarCoder fine-tuned for text-to-SQL. Very accurate.	15B:Q4_K_M	—
duckdb-nsql	7B	SQL (DuckDB)	Text-to-SQL specifically for DuckDB analytics.	7B:Q6_K	—
nuextract	3.8B	Info Extraction	Phi-3 fine-tuned on synthetic extraction data. Structured data from documents.	3.8B:Q8_0	3.8B:Q4_K_M
bespoke-minicheck	7B	Fact-checking	SOTA fact-checking model by Bespoke Labs.	7B:Q5_K_M	—
llama3-chatqa	8B/70B	RAG, Conv QA	NVIDIA Llama 3 optimized for conversational QA and RAG pipelines.	8B:Q6_K	—
llama-guard3	1B/8B	Content Safety	Meta content safety classifier for LLM inputs and outputs.	8B:Q4_K_M	1B:Q4_K_M
samantha-mistral	7B	Companion	Companion assistant trained in philosophy, psychology, and personal relationships.	7B:Q5_K_M	—

4. Platform Recommendations Summary

4.1 CachyOS — Top Picks by Use Case

CachyOS with GPU acceleration can run most models in the library. Recommendations below assume at least 8–16GB VRAM. Adjust quantization down (Q4_K_M) for tighter VRAM.

Use Case	Primary Recommendation	Runner-Up	Pull Command
Daily General Chat	llama3.1:8b	qwen2.5:7b	ollama run llama3.1:8b
Heavy Reasoning	deepseek-r1:32b	qwq:32b	ollama run deepseek-r1:32b
Code Generation	qwen2.5-coder:14b	devstral:24b	ollama run qwen2.5-coder:14b
Agentic / Tool Use	llama3.3:70b	hermes3:8b	ollama run llama3.3:70b
Vision / Image Input	llama3.2-vision:11b	gemma3:12b	ollama run llama3.2-vision:11b
RAG Embeddings	nomic-embed-text	bge-m3	ollama pull nomic-embed-text
Creative / Uncensored	dolphin3:8b	dolphin-mixtral:8x7b	ollama run dolphin3
Long Context (128K+)	mistral-nemo:12b	qwen2.5:14b	ollama run mistral-nemo
Math / Science	deepseek-r1:14b	qwen2-math:7b	ollama run deepseek-r1:14b
SQL / Databases	sqlcoder:15b	duckdb-nsql:7b	ollama run sqlcoder:15b
Document Extraction	nuextract:3.8b	deepseek-ocr:3b	ollama run nuextract
Translation (55 langs)	translategemma:12b	aya-expanse:8b	ollama run translategemma:12b
Medical Queries	meditron:7b	medllama2:7b	ollama run meditron

4.2 Pixel 9 / Android (Termux) — Top Picks

On the Pixel 9, Ollama runs CPU-only. With ~5–8GB available RAM in Termux, models up to 3B at Q4_K_M are practical. The 3.8B Phi and 4B Gemma3 work with patience.

⚡ **Speed expectation:** CPU inference yields roughly 3–8 tokens/sec for 1B models and 1–3 tokens/sec for 3B. Best for offline assistant tasks, not real-time high-volume use.

Use Case	Best Model	Tag	Why
Best All-Rounder	llama3.2:3b	3b-instruct-q4_K_M	Tool calling, instruction following, Meta quality at 3B. Best balance.
Fastest Response	llama3.2:1b	1b-instruct-q4_K_M	1B loads in seconds. Quick lookups and simple tasks.
Reasoning	deepseek-r1:1.5b	1.5b-q4_K_M	Thinking mode in 1.5B. Chain-of-thought on mobile.
Math / Logic	deepscaler:1.5b	1.5b-q4_K_M	Beats o1-preview on math evaluations at just 1.5B.
Vision (Mobile)	moondream:1.8b	1.8b-q4_K_M	Designed for edge/mobile. Smallest usable vision model.
Code Assist	qwen2.5-coder:3b	3b-q4_K_M	Best 3B coding model. Code generation and debugging in Termux.
Tiny Agentic	smollm2:1.7b	1.7b-q4_K_M	1.7B with tool calling. Minimal footprint for automated tasks.
Best Quality ~4B	phi4-mini:3.8b	3.8b-q4_K_M	Microsoft Phi-4 Mini. Function calling and multilingual. Best quality near 4B.
Absolute Minimum	tinyllama:1.1b	1.1b-q4_K_M	1.1B, loads instantly, works in under 3GB RAM.
Thinking + Tiny	lrm2.5-thinking:1.2b	1.2b-q4_K_M	Hybrid reasoning in 1.2B. On-device thinking mode.

5. CLI & API Reference

5.1 Core Commands

```
ollama pull <model>:<tag>      # Download a model
ollama run <model>              # Interactive REPL (pulls first if needed)
ollama list                    # List downloaded models
ollama rm <model>              # Delete a model
ollama show <model>            # Show model info and parameters
ollama ps                     # Show currently loaded models
ollama cp <src> <dst>          # Copy/rename a model
ollama serve                   # Start server (foreground)
ollama version                 # Show Ollama version
```

5.2 Pulling Specific Quantizations

```
# Default (usually Q4_K_M)
ollama pull llama3.1:8b

# Specific quantization
ollama pull llama3.1:8b-instruct-q5_K_M
ollama pull deepseek-r1:7b-qwen-distill-q8_0
ollama pull qwen2.5-coder:14b-instruct-q4_K_M

# Browse all available tags:
# https://ollama.com/library/<model>/tags
```

5.3 REST API

```
# OpenAI-compatible chat completion
curl http://localhost:11434/v1/chat/completions \
  -H 'Content-Type: application/json' \
  -d '{"model":"llama3.1","messages":[{"role":"user","content":"Hello"}]}'

# Raw generate (streaming off)
curl http://localhost:11434/api/generate \
  -d '{"model":"llama3.1","prompt":"Why is the sky blue?","stream":false}'

# Embeddings
curl http://localhost:11434/api/embeddings \
  -d '{"model":"nomic-embed-text","prompt":"search query here"}'

# List models
curl http://localhost:11434/api/tags
```

5.4 Modelfile — Custom Configuration

```
# ~/Modelfile
FROM llama3.1:8b

SYSTEM """
You are a CachyOS Linux technical assistant.
Always provide shell commands when relevant.
"""

PARAMETER temperature 0.7
PARAMETER top_p 0.9
```

```
PARAMETER num_ctx 8192
PARAMETER num_predict 2048

ollama create myassistant -f ~/Modelfile
ollama run myassistant
```

5.5 CachyOS Shell Aliases

```
# Add to ~/.bashrc or ~/.zshrc
alias ols='ollama serve'
alias olr='ollama run'
alias oll='ollama list'
alias olp='ollama pull'
alias olps='ollama ps'
alias olstart='sudo systemctl start ollama.service'
alias olstop='sudo systemctl stop ollama.service'
alias ollogs='journalctl -fu ollama'
alias chat='ollama run llama3.1:8b'
alias codex='ollama run qwen2.5-coder:14b'
alias think='ollama run deepseek-r1:32b'
```

5.6 Troubleshooting

CachyOS

- **Model runs on CPU instead of GPU:** Confirm CUDA/ROCm package installed. Run `ollama ps` to check GPU layer count shown as 'offset'.
- **OOM / doesn't fit VRAM:** Use a smaller quantization tag or add `OLLAMA_KV_CACHE_TYPE=q4_0` to the systemd override.
- **Service won't start:** Check `journalctl -fu ollama`. Ensure nothing else is on port 11434.
- **ROCm model still on CPU:** Add user to render and video groups: `sudo usermod -aG render,video $USER`, then logout/login.

Pixel 9 / Termux

- **Ollama killed mid-inference:** Lock Termux in recents and disable battery optimization in Android settings.
- **Command not found after restart:** Re-run `pkg install ollama` or check `$PATH` includes Termux bin directory.
- **Download stalls on large models:** Use `export OLLAMA_HTTP_CLIENT="curl -L --retry 3"` before pulling.
- **Out of memory:** Set `OLLAMA_MAX_LOADED_MODELS=1` and `OLLAMA_NUM_PARALLEL=1`. Drop to `Q3_K_M` if needed.

6. Complete Alphabetical Model Index

Quick-reference index of every model with pull counts, category, and platform picks. ★ indicates a top recommended pick for that platform.

Model	Pulls	Category	Key Capabilities	CachyOS	Pixel 9
all-minilm	2.5M	Embedding	Tiny sentence embeddings	22M:Q8_0	22M ★
aya	461K	Multilingual	23 languages, conversational	8B:Q6_K	—
aya-expanse	385K	Multilingual, Tools	23 languages, improved	8B:Q5_K_M	—
baklava	534K	Vision	Mistral + LLaVA architecture	7B:Q5_K_M	—
bespoke-minicheck	218K	Fact-checking	SOTA fact verification	7B:Q5_K_M	—
bge-m3	3.4M	Embedding	Multilingual, multi-granularity	567M	—
codegemma	2.3M	Code, FIM	Fill-in-middle, Google	7B:Q5_K_M	2B:Q4_K_M
codellama	4.3M	Code	Meta, 20+ languages	34B:Q4_K_M	—
codestral	849K	Code Generation	Mistral code specialist	22B:Q4_K_M	—
codeqwen	480K	Code	64K ctx, 92 languages	7B:Q6_K	—
cogito	1.5M	Reasoning, Tools	Hybrid reasoning, outperforms Llama	14B:Q5_K_M	3B:Q4_K_M
command-r	739K	Tools, RAG	Cohere, long context	35B:Q4_K_M	—
deepcoder	666K	Code, Reasoning	Open 14B at O3-mini level	14B:Q4_K_M	1.5B:Q4_K_M ★
deepscaler	1.1M	Math Reasoning	Beats o1-preview at 1.5B	1.5B:Q8_0	1.5B:Q4_K_M ★
deepseek-coder	3.1M	Code	2T code tokens trained	6.7B:Q6_K	1.3B:Q4_K_M
deepseek-coder-v2	1.7M	Code (MoE)	GPT-4-Turbo code parity	16B:Q4_K_M	—
deepseek-r1	78.8M	Reasoning, Thinking	Approaches O3/Gemini 2.5 Pro	32B:Q4_K_M	1.5B:Q4_K_M ★
deepseek-v3	3.5M	MoE, General	671B total, 37B active	40GB+ needed	—

Model	Pulls	Category	Key Capabilities	CachyOS	Pixel 9
devstral	772K	Code Agents	Best open coding agent	24B:Q4_K_M	—
dolphin3	3.6M	General, Uncensored	Ultimate general local model	8B:Q5_K_M	—
dolphin-llama3	1.1M	Instruct, Code	Dolphin on Llama 3	8B:Q6_K	—
dolphin-mixtral	1.1M	MoE, Code, Uncensored	Uncensored Mixtral MoE	8x7B:Q4_K_M	—
dolphin-phi	1.2M	Compact, Uncensored	2.7B Phi-based tiny uncensored	2.7B:Q8_0	—
exaone-deep	506K	Reasoning, Math	LG AI, 7.8B beats o1-mini	7.8B:Q5_K_M	2.4B:Q4_K_M ★
falcon3	2.2M	General	Efficient under 10B params	7B:Q5_K_M	1B:Q4_K_M
gemma	6M	General	Google, original series	7B:Q5_K_M	2B:Q4_K_M
gemma2	16.6M	General	Google Gemma 2, flash attention	9B:Q6_K	2B:Q4_K_M
gemma3	32.4M	Vision, General	Gemini-class, multi-image input	12B:Q5_K_M	4B:Q4_K_M ★
glm4	492K	Multilingual, Tools	Competitive with Llama 3	9B:Q5_K_M	—
gpt-oss	7.2M	Tools, Thinking	OpenAI open-weight reasoning	20B:Q4_K_M	—
granite-code	710K	Code (IBM)	Enterprise code intelligence	20B:Q4_K_M	3B:Q4_K_M
granite3.1-moe	2.3M	Tools, MoE	IBM tiny MoE, low latency	3B:Q8_0	1B:Q4_K_M
granite4	749K	Tools, Edge	IBM edge models, tool calling	3B:Q8_0	1B:Q4_K_M ★
hermes3	653K	Tools, Agentic	Nous Research, function calling	8B:Q5_K_M	3B:Q4_K_M
l1m2.5-thinking	833K	Tools, Thinking, Edge	On-device thinking in 1.2B	1.2B:Q8_0	1.2B:Q4_K_M ★
llama-guard3	393K	Content Safety	Meta content classifier	8B:Q4_K_M	1B:Q4_K_M
llama2	5.5M	General	Meta Llama 2	13B:Q4_K_M	—
llama2-uncensored	1.8M	Uncensored	Uncensored Llama 2	7B:Q6_K	—

Model	Pulls	Category	Key Capabilities	CachyOS	Pixel 9
llama3	16.6M	General	Meta Llama 3	8B:Q5_K_M	—
llama3.1	110.6M	Tools, General	Meta flagship, 128K context	70B:Q4_K_M	—
llama3.2	58.5M	Tools, Mobile	Meta compact on-device	3B:Q8_0	3B:Q4_K_M ★
llama3.2-vision	3.9M	Vision	Llama 3.2 with image input	11B:Q4_K_M	—
llama3.3	3.3M	Tools, SOTA 70B	New Meta SOTA, = 405B quality	70B:Q4_K_M	—
llava	12.9M	Vision	Classic LLaVA 1.6, general VQA	13B:Q4_K_M	—
llava-llama3	2.1M	Vision	LLaVA on Llama 3 Instruct	8B:Q5_K_M	—
llava-phi3	201K	Vision, Compact	LLaVA from Phi-3 Mini	3.8B:Q8_0	3.8B:Q4_K_M ★
magistral	1.1M	Reasoning, Tools	Mistral reasoning 24B	24B:Q4_K_M	—
mathstral	234K	Math, Science	Mistral math specialist	7B:Q6_K	—
medllama2	254K	Medical	Llama 2 for medical Q&A	7B:Q5_K_M	—
meditron	315K	Medical	Llama 2 for clinical reasoning	7B:Q5_K_M	—
minicpm-v	4.6M	Vision	Vision-language multimodal	8B:Q5_K_M	—
ministral-3	508K	Tools, Vision, Edge	Mistral edge model	3B:Q5_K_M	3B:Q4_K_M
mistral	25.8M	Tools, General	Mistral 7B v0.3, very fast	7B:Q6_K	—
mistral-large	576K	Multilingual, Code	Mistral flagship 123B, 128K	64GB+ needed	—
mistral-nemo	3.4M	Tools, 128K ctx	12B with huge context	12B:Q4_K_M	—
mistral-small	2.4M	Tools, General	Mistral Small 3, sub-70B SOTA	24B:Q4_K_M	—
mixtral	1.9M	Tools, MoE	Mistral MoE, coding excellence	8x7B:Q5_K_M	—
moondream	662K	Vision, Edge	Designed for mobile/CPU	1.8B:Q8_0	1.8B:Q4_K_M ★

Model	Pulls	Category	Key Capabilities	CachyOS	Pixel 9
mxbai-embed-large	7.7M	Embedding	SOTA retrieval embeddings	335M	—
nemotron-mini	290K	Tools, RAG	NVIDIA 4B for RAG and functions	4B:Q8_0	4B:Q4_K_M
nomic-embed-text	55.2M	Embedding	Best general RAG embedding	137M	—
nuextract	220K	Info Extraction	Phi-3 for structured extraction	3.8B:Q8_0	3.8B:Q4_K_M
olmo2	3.5M	General, Open	AllenAI, fully open weights	13B:Q5_K_M	—
opencoder	429K	Code	Open reproducible code LLM	8B:Q5_K_M	1.5B:Q4_K_M
openthinker	808K	Reasoning	Open DeepSeek-R1 distill	32B:Q4_K_M	—
phi	1.1M	General	Microsoft Phi-2, 2.7B	2.7B:Q8_0	2.7B:Q4_K_M
phi3	16.1M	General	Microsoft Phi-3 Mini/Medium	14B:Q4_K_M	3.8B:Q4_K_M ★
phi3.5	566K	General	Updated Phi-3	3.8B:Q8_0	3.8B:Q4_K_M
phi4	7.2M	Reasoning, Code	Microsoft SOTA 14B	14B:Q5_K_M	—
phi4-mini	915K	Tools, Math, Multilingual	Phi-4 compact with tools	3.8B:Q8_0	3.8B:Q4_K_M ★
phi4-mini-reasoning	157K	Reasoning, Math	Phi-4 Mini for math steps	3.8B:Q8_0	3.8B:Q4_K_M ★
phi4-reasoning	1.2M	Reasoning, Math	Phi-4 14B reasoning specialist	14B:Q4_K_M	—
qwq	2.1M	Reasoning	Qwen reasoning, strong math	32B:Q4_K_M	—
qwen	5.6M	General	Qwen 1.5 series	14B:Q4_K_M	0.5B
qwen2	4.9M	Tools, General	Qwen 2 series	7B:Q6_K	0.5B:Q8_0
qwen2-math	467K	Math	Qwen2 math specialist	7B:Q5_K_M	1.5B:Q4_K_M
qwen2.5	22.4M	Tools, 128K, Multilingual	Alibaba workhorse 18T tokens	14B:Q5_K_M	3B:Q4_K_M
qwen2.5-coder	11.4M	Code, Tools	Best code model under 14B	14B:Q5_K_M	3B:Q4_K_M ★

Model	Pulls	Category	Key Capabilities	CachyOS	Pixel 9
qwen3	22.4M	Tools, Thinking, MoE	Latest Qwen, dense+MoE	14B:Q5_K_M	1.7B:Q4_K_M ★
reader-lm	355K	HTML→Markdown	Converts web content to MD	1.5B:Q8_0	0.5B:Q8_0
sailor2	174K	SE Asian Languages	Multilingual SE Asia	8B:Q5_K_M	1B:Q4_K_M
samantha-mistral	404K	Companion	Philosophy/psychology companion	7B:Q5_K_M	—
smollm	993K	Compact	HuggingFace tiny family	1.7B:Q8_0	1.7B:Q4_K_M
smollm2	2.6M	Tools, Compact	SmolLM2 with tool calling	1.7B:Q8_0	1.7B:Q4_K_M ★
sqlcoder	438K	SQL Code	StarCoder for SQL generation	15B:Q4_K_M	—
stable-code	450K	Code, Compact	3B = CodeLlama 7B quality	3B:Q8_0	3B:Q4_K_M ★
starcoder2	2.1M	Code	Next-gen StarCoder, transparent	15B:Q5_K_M	3B:Q4_K_M
tinydolphin	364K	Compact, Uncensored	Dolphin data on TinyLlama	1.1B:Q8_0	1.1B:Q4_K_M
tinyllama	3.7M	Compact	1.1B trained on 3T tokens	1.1B:Q8_0	1.1B:Q4_K_M ★
translategemma	442K	Translation (55 langs)	Gemma 3-based translation	12B:Q4_K_M	4B:Q4_K_M
wizard-vicuna-uncensored	567K	Uncensored	Wizard Vicuna uncensored	13B:Q4_K_M	—
wizardlm2	680K	Chat, Multilingual, Reasoning	Microsoft WizardLM 2	7B:Q5_K_M	—
yi	605K	Bilingual (Chinese/English)	Yi 1.5 bilingual	9B:Q5_K_M	—
yi-coder	431K	Code	Yi code, SOTA under 10B	9B:Q5_K_M	1.5B:Q4_K_M
zephyr	585K	Assistant	Mistral fine-tuned for chat	7B:Q6_K	—